

Daniel Alexander Villarruel Yánez

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Education

MSc in Physics (CGPA 98%) 2025 – Present

School of Physical Sciences and Nanotechnology, Yachay Tech, Urcuquí – Ecuador

- Thesis: *Magnetohydrodynamics of Wind-Cloud Systems in Starburst Galaxies*
Supervisor: Dr. Wladimir Banda-Barragán.
- Expected Graduation: February 2027.

BSc in Physics (5-year programme, CGPA 91%) 2020 – 2025

School of Physical Sciences and Nanotechnology, Yachay Tech, Urcuquí – Ecuador

- Thesis: *Starburst Heating and Spectra in Galactic Winds*
Supervisor: Dr. Wladimir Banda-Barragán.
- Specialisation in Computational Physics for Astrophysics and Materials Science.

Research Experience

Research Intern 2026 – Present

Max Planck Institute for Astronomy, Heidelberg, Germany

- Project: Energy budget in asymmetric disk outflows through numerical simulations
Supervisor: Dr. Giancarlo Mattia
- Running and post-processing 2.5D MHD disk-jet simulations using PLUTO.
- Implementing and testing heating and cooling prescriptions for disk outflows.

Graduate Researcher 2025 – Present

Computational Physics Plus CPhys+ Group, Yachay Tech, Urcuquí – Ecuador

- Lead developer of `py4radiation`, a python package that incorporates photoionisation and radiative heating and cooling rates into hydrodynamical simulations of gas in galaxies.
- Running and post-processing HD/MHD simulations using PLUTO, Trident, Starburst99, and Cloudy.
- Leading code development, parallelisation and HPC deployments.

Research Assistant Jul – Dec 2024

School of Physical Sciences and Nanotechnology, Yachay Tech, Urcuquí – Ecuador

- Project: *Prediction of thermoelectric properties of layered oxide $\text{Ca}_3\text{Co}_4\text{O}_9$ from first-principles*
Supervisor: Dr. Henry P. Pinto
- Led development of DFTB+ workflows for first-principles thermodynamical calculations.

Programming Skills

Languages

Python, C/C++, Julia

Scientific Computing and HPC

MPI, OpenMP, SLURM, SuperMUC-NG, CUDA

Astrophysics

Starburst99, Cloudy, yt, Trident, PLUTO, gPLUTO, Illustris TNG, GNU Astro

Publications

- **Villarruel, D.**, Banda-Barragán, W., Casavecchia, B. "Starburst heating and synthetic ion column densities in multiphase galactic outflows". *Proceedings of the International Astronomical Union* (in press), 2024. DOI
- Banda-Barragán, W., Antipov, A., **Villarruel, D.** "Survival and synthetic observables of neutral atomic hydrogen in galactic wind simulations". *Proceedings of the International Astronomical Union* (in press), 2024. DOI
- **Villarruel, D.**, Banda-Barragán, W., Casavecchia, B. "py4radiation: linking radiation to hydrodynamical simulations of gas in galaxies" (in preparation).

Conferences, Workshops, and Research Schools

- 46th International School for Young Astronomers (2025) – Project "Surface brightness profiles of Ultra Deep Field galaxies using HST data".
- XXXII International Astronomical Union General Assembly (2024) – Poster (FM2) "Starburst heating and synthetic ion column densities in multiphase galactic outflows".

Teaching Experience

Lecturer 2024 – Present

The Ecuadorian Olympiads on Astronomy and Astrophysics

- Preparing and delivering workshops and lectures in Mathematics, Physics, and Astrophysics for 400+ talented high school students participating in international Astronomy competitions.
- Planning, executing, and evaluating the scientific programme at the national and international levels.

Teaching Assistant (Quantum Mechanics I) Jan – Jun 2024

School of Physical Sciences and Nanotechnology, Yachay Tech, Urcuquí – Ecuador

- Planned and delivered weekly tutorials and problem solving sessions for 23 students.
- Graded assignments, quizzes, and exams.

Service and Leadership

President 2021 – Present

The Ecuadorian Olympiads on Astronomy and Astrophysics

- Leading the non-profit foundation to administer the National Astronomy and Physics Olympiads.
- Coordinating mentors, trainers, and students for the scientific programme of the olympiads.
- Obtained a 1000 USD travel grant from the Organisation of Ibero-American States (OEI).

National Delegate representing Ecuador 2021 – Present

The Ecuadorian Olympiads on Astronomy and Astrophysics

- Team Leader at the International Olympiad on Astronomy and Astrophysics (IOAA) and the Latin American Olympiad on Astronomy and Astronautics (OLAA).
- Lead 8 delegations of students to international competitions.
- Managed logistics, travel itineraries and budgets across the Americas, Europe, and Asia.

Languages

Spanish (native), English (C1), German (B2)